

Exp 5.1

→

Develop a recursive and non recursive function fact to find the factorial of a number n ? defined by $FACT(n) = 1$, if $n = 0$. Otherwise, $FACT(n) = n * FACT(n-1)$. Using this function, write a C program to compute the binomial coefficient calculate the result for different values of n and r with suitable messages

→

```
#include <stdio.h>
```

```
int fact_recursive (int n)
```

```
{
```

```
    if (n == 0)
```

```
        return 1;
```

```
    else
```

```
        return n * fact_recursive (n-1);
```

```
}
```

```
int fact_non_recursive (int n)
```

```
{
```

```
    int fact = 1;
```

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```
for (int i = 1 ; i <= n ; i++)
```

```
{
```

```
    fact *= i ;
```

```
}
```

```
return fact ;
```

```
}
```

```
int binomial_recursive (int n , int r)
```

```
{
```

```
return fact_recursive (n) / fact_recursive (r) * fact_recursive (n-r)
```

```
}
```

```
int binomial_non_recursive (int n , int r)
```

```
{
```

```
return fact_non_recursive (n) / fact_non_recursive (r) * fact_non_recursive (n-r)
```

```
}
```

```
int main () {
```

```
    int n , r ;
```

```
    printf (" Enter n : " ) ;
```

```
    scanf ("%d", &n) ;
```

```
    printf (" Enter r : " ) ;
```

```
    scanf ("%d", &r) ;
```

```
if (r > n || n < 0 || r < 0) {  
    printf("Invalid input! r must be <= n and  
    both >= 0. \n");  
    return 0;  
}
```

```
printf("\n Results in Tabular form : \n");  
printf("-----\n");  
printf(" n | r | C(n, r) Recursive | C(n, r) non recursive \n");  
printf("-----\n");
```

```
for (int i = 0; i <= n; i++)
```

```
{  
    for (int j = 0; j <= i; j++)
```

```
        printf("%12d | %12d | %12d | %12d \n", i, j,  
        binomial_recursive(i, j), binomial_non_recursive(i, j));  
    }
```

```
}
```

```
printf("-----");
```

```
return 0;
```

```
}
```


Exp = 5.2

→

Develop a recursive function GCD (num1, num2) that accepts two integer arguments. Write a C program that makes this function to find the greatest common divisor of two given integers

→

include <stdio.h>

int GCD (int num1, int num2)

{

if (num 2 == 0)

return num 1 ;

else

return GCD (num2, num1 % num2);

}

int main ()

{

int a, b ;

printf (" Enter first number : ");

scanf ("%d", &a);

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```
printf("Enter second number : ");  
scanf("%d", &b);
```

```
if (a <= 0 || b <= 0)
```

```
{  
    printf("Please enter positive integers only\n");  
    return 0;  
}
```

```
printf("gcd of %d & %d is: %d\n", a, b, GCD(a,b));
```

```
return 0;  
}
```

exp_5.c > C exp2.c > ...

```
1  #include <stdio.h>
2
3  int GCD(int num1, int num2)
4  {
5      if (num2 == 0)
6          return num1;
7      else
8          return GCD(num2, num1 % num2);
9  }
10
11 int main()
12 {
13     int a, b;
14
15     printf("Enter first number: ");
16     scanf("%d", &a);
17
18     printf("Enter second number: ");
19     scanf("%d", &b);
20
21     if (a <= 0 || b <= 0)
22     {
23         printf("Please enter positive integers only.\n");
24         return 0;
25     }
26
27     printf("GCD of %d and %d is: %d\n", a, b, GCD(a, b));
28
29     return 0;
30 }
31
```

PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_5.c\output'

PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_5.c\output> & .\exp2.exe

Enter first number: 48

Enter second number: 18

GCD of 48 and 18 is: 6

PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_5.c\output> |

Exp = 5.3

→

Develop a recursive function FIBO (num) that accepts an integer argument. Write a C program that invokes this function to generate the fibonacci sequence up to num

→

include <stdio.h>

int fibonacci (int num)

{

if (num == 0)

return 0;

else if (num == 1)

return 1;

else

return fibonacci (num-1) + fibonacci (num-2)

}

int main ()

{

int n;

printf ("Enter the number of terms : ");

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```
scanf ("%d", &n);
```

```
if (n <= 0)
```

```
{
    printf ("Please enter a positive integer - \n");
    return 0;
}
```

```
printf ("fibonacci sequence up to %d terms: \n", n)
```

```
for (int i = 0; i < n; i++)
```

```
{
    printf ("%d", fibonacci (i));
}
```

```
printf ("\n");
```

```
return 0;
```

```
}
```

```

c exp2.c exp_5.c  exp3.c exp_5.c  exp4.c exp_5.c  exp5.c
exp_5.c > C exp3.c > ...
1  #include <stdio.h>
2
3  int fibonnaci(int num)
4  {
5      if (num == 0)
6          return 0;
7      else if (num == 1)
8          return 1;
9      else
10         return fibonnaci(num - 1) + fibonnaci(num - 2);
11 }
12
13 int main()
14 {
15     int n;
16
17     printf("Enter the number of terms: ");
18     scanf("%d", &n);
19
20     if (n <= 0)
21     {
22         printf("Please enter a positive integer.\n");
23         return 0;
24     }
25
26     printf("fibonnacinacci sequence up to %d terms:\n", n);
27
28     for (int i = 0; i < n; i++)
29     {
30         printf("%d ", fibonnaci(i));
31     }
32
33     printf("\n");
34
35     return 0;
36 }

```

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  C/C++ Compile Run
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_5.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_5.c\output> & .\exp3.exe
Enter the number of terms: 10
fibonnacinacci sequence up to 10 terms:
0 1 1 2 3 5 8 13 21 34
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_5.c\output>

```

Exp = 5.4

→

Develop a C function `ISPRIME (num)` that accepts an integer argument and returns 1 if the argument is prime or otherwise. Write a C program that invokes this function to generate prime numbers between the given ranges.

→

```
# include <stdio.h>
```

```
int ISPRIME (int num)
```

```
{
```

```
    if (num <= 1)
```

```
        return 0;
```

```
    for (int i = 2; i <= num-1; i++)
```

```
    {
```

```
        if (num % i == 0)
```

```
            return 0;
```

```
    }
```

```
    return 1;
```

{

int main ()

{

int lower, upper;

printf ("Enter the lower range: ");

scanf ("%d", &lower);

printf ("Enter the upper range: ");

scanf ("%d", &upper);

if (lower > upper) {

printf ("Invalid range! lower should be $\leq$ upper
.\n");

return 0;

}

printf ("Prime numbers between %d and %d
are: \n", lower, upper);for (int num = lower; num $\leq$ upper; num++)

{

if (ISPRIME(num))

{

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```
printf ("%d", num);
```

```
}
```

```
}
```

```
}
```

```
printf ("\n");
```

```
return 0;
```

```
}
```

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```

c > C exp4.c > main()
#include <stdio.h>

int ISPRIME(int num)
{
    if (num <= 1)
        return 0;

    for (int i = 2; i <= num-1; i++)
    {
        if (num % i == 0)
            return 0;
    }

    return 1;
}

int main()
{
    int lower, upper;

    printf("Enter the lower range: ");
    scanf("%d", &lower);

    printf("Enter the upper range: ");
    scanf("%d", &upper);

    if (lower > upper) {
        printf("Invalid range! Lower should be <= Upper.\n");
        return 0;
    }

    printf("Prime numbers between %d and %d are:\n", lower, upper);

    for (int num = lower; num <= upper; num++)
    {
        if (ISPRIME(num))
        {
            printf("%d ", num);
        }
    }

    printf("\n");

    return 0;
}

```

```

PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_5.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_5.c\output> & .\exp4.exe
Enter the lower range: 10
Enter the upper range: 30
Prime numbers between 10 and 30 are:
11 13 17 19 23 29
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_5.c\output>

```

Exp = 5.5

→

Develop a function REVERSE (str) that accepts a string argument. Write a C program that invokes this function to find the reverse of a given string

→

include <stdio.h>

include <string.h>

void REVERSE (char str [])

{

int length, start, end;

char temp;

length = strlen (str);

start = 0;

end = length - 1;

while (start < end)

{

temp = str [start];

str [start] = str [end];

str [end] = temp;

```
start ++ ;
```

```
end -- ;
```

```
}
```

```
}
```

```
int main ()
```

```
{
```

```
char str [100];
```

```
printf (" enter a string: ");
```

```
fgets (str, sizeof(str), stdin);
```

```
str [strcspn (str, "\n")] = '\0';
```

```
REVERSE (str);
```

```
printf (" Reversed string : %s\n", str);
```

```
return 0
```

```
}
```

